

CSE-350 Quiz-4

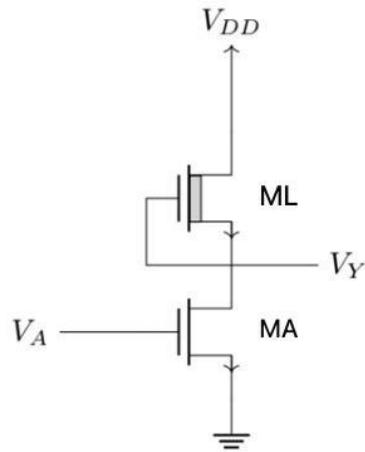
NMOS and CMOS Logic

Set-A

Total – 20 marks

Time – 20 minutes

Name -	Id -	Section -
Problem-1		10 marks



A typical NMOS Inverter with depletion type load

It is given that $K_n' = 100 \mu\text{A}/\text{V}^2$, $V_{TNL} = -0.6 \text{ V}$, $V_{TNA} = 0.4 \text{ V}$, $(W/L)_L = 4$, $(W/L)_A = 1$

Note: $K_n = K_n' (W/L)$

(a)	If $V_{DD} = 5\text{V}$, find V_Y and the current through the MOSFETs M_L and M_A for all input cases.	8
(b)	Find the average power dissipation.	2

Problem - 1 Soln:

$$(a) \quad K_n = K_n' (W/L)$$

Driver, M_A $\rightarrow K_{nA} = 100 \times 1 \mu A/V^2, V_{TnA} = 0.4V$
 $V_{GS_A} = V_A, V_{DS_A} = V_Y$

Load, M_L $\rightarrow K_{nL} = 100 \times 4 = 400 \mu A/V^2, V_{TnL} = -0.6V$
 $V_{GS_L} = 0V, V_{DS_L} = V_{DD} - V_Y$

Case-1: $V_A = 0V \rightarrow M_A: \text{cutoff}, M_L: \text{Linear/Triode}$
 $\hookrightarrow I_{DA} = 0 \quad \hookrightarrow I_L = I_L (\text{triode})$

here, $I_L = I_{DA}$

$$\Rightarrow K_{nL} \{ 2(V_{GS_L} - V_{TnL}) V_{DS_L} - V_{DS_L}^2 \} = 0$$

$$\Rightarrow 400 \mu A/V^2 \{ 2(0 + 0.6)(V_{DD} - V_Y) - (V_{DD} - V_Y)^2 \} = 0$$

$$\Rightarrow 1.2(5 - V_Y) - (5 - V_Y)^2 = 0$$

$$\Rightarrow (5 - V_Y) \{ 1.2 - (5 - V_Y) \} = 0$$

Solving yields $\rightarrow V_Y = 5V, 3.8V$

no current $\rightarrow$ no drop $\therefore V_Y = 3.8$ not acceptable

$$\therefore \boxed{V_Y = 5V}$$

$$\therefore I_D = I_L = 0 \text{ mA}$$

Case-2: $V_A = 5V \rightarrow M_A: \text{Linear/Triode}, M_L: \text{Sat.}$

$$\hookrightarrow I_{DA} = I_D (\text{triode}), \hookrightarrow I_L = I_D (\text{sat.})$$

$$I_{DA} = I_L$$

$$\Rightarrow K_{nA} \{ 2(V_{GS_A} - V_{TnA}) V_{DS_A} - V_{DS_A}^2 \} = K_{nL} (V_{GS_L} - V_{TnL})^2$$

$$\Rightarrow 100 \{ 2(V_Y - 0.4) V_Y - V_Y^2 \} = 400 (0 + 0.6)^2$$

$$\Rightarrow 2(5 - 0.4) V_Y - V_Y^2 = 4 \times (0.6)^2$$

Solving $\rightarrow V_Y = 9.04V \times, 0.156V \checkmark$

$$\boxed{V_Y = 0.156V}$$

$$I_{DA} = I_L = 400 \times 10^{-6} (0 + 0.6)^2 = 0.144 \text{ mA}$$

$$(b) \quad P_1 = (5 - 0) \times 0 = 0, \quad P_2 = (5 - 0) \times 0.144 = 0.72 \text{ mW}$$

$$\therefore P_{\text{Avg}} = \frac{P_1 + P_2}{2} = 0.36 \text{ mW}$$

Problem-2

10 marks

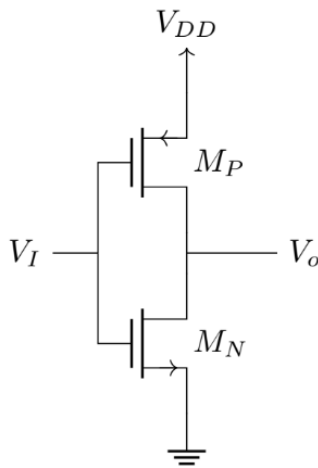


Figure 1 - CMOS Inverter

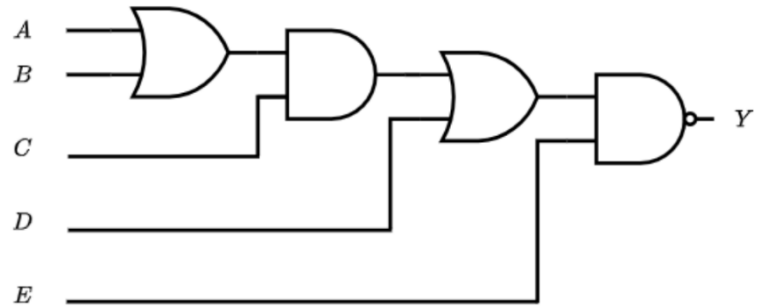


Figure 2

It is given that $V_{DD} = 5\text{ V}$, $V_{TN} = -V_{TP} = 0.5\text{ V}$, $K_N = K_P = 10\text{ }\mu\text{A/V}^2$

(a)	Draw and label the VTC of the above CMOS Inverter shown in Figure 1 and then determine the values of critical voltages V_{It} , V_{ONt} , and V_{OPt}	5
(b)	Implement the Boolean logic Y shown in Figure 2 using a CMOS logic circuit	5

Problem-2

10 marks

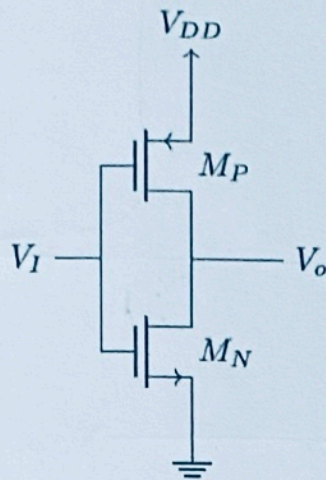


Figure 1 - CMOS Inverter

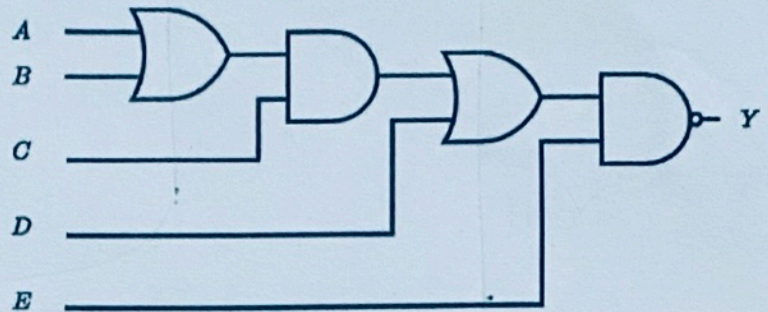


Figure 2

It is given that $V_{DD} = 5\text{ V}$, $V_{TN} = -V_{TP} = 0.5\text{ V}$, $K_N = K_P = 10\text{ }\mu\text{A/V}^2$

(a)	Draw and label the VTC of the above CMOS Inverter shown in Figure 1 and then determine the values of critical voltages V_{IL} , V_{ONL} , and V_{OPL}	5
(b)	Implement the Boolean logic Y shown in Figure 2 using a CMOS logic circuit	5

$$a) \quad V_{IL} = \frac{\sqrt{K_P}(V_{DD} + V_{TP}) + \sqrt{K_N} \cdot V_{TN}}{\sqrt{K_N} + \sqrt{K_P}}$$

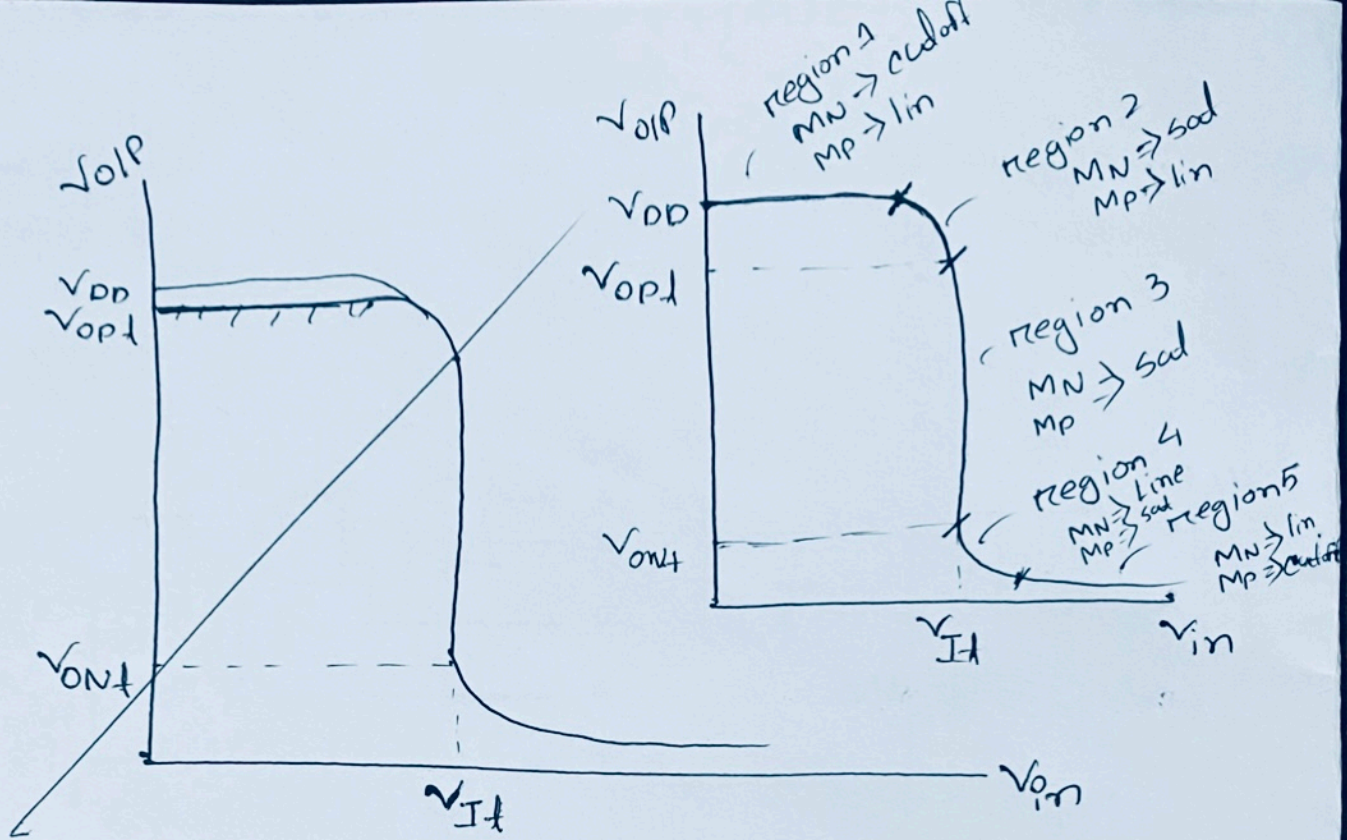
$$= 2.5\text{ V}$$

$$V_{ONL} = V_{IL} + |V_{TP}|$$

$$= 3\text{ V}$$

$$V_{OPL} = V_{IL} - V_{TN}$$

$$= 2\text{ V}$$



b) $\overline{AB((A+B)C+D)E}$

